Perturbations to the 30B-10 Cylinder Criticality Control System

Zachary Petersen[1], Aly Gaylard[2]
Alex Lang[3], Cihangir Celik[3], Walid A Metwally[3]
[1]*University of Wisconsin-Madison, Madison, Wisconsin, USA*
[2]*Texas A&M University, College Station, USA*
[3]*Oak Ridge National Laboratory Nuclear Energy and Fuel Cycle Division, Oak Ridge, Tennessee, USA*

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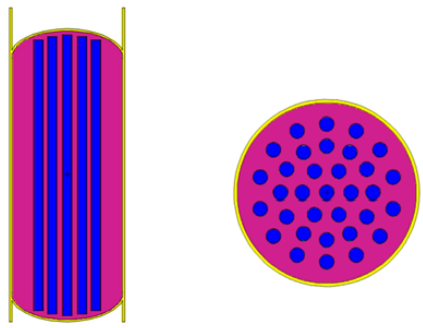
INTRODUCTION

The 30B-10 is an advanced canister designed to transport fresh fuel. This innovative canister incorporates high-assay low enriched uranium (HALEU) fuel that is currently used in advanced nuclear reactors. The 30B-10 cylinder is able to accommodate up to 10% enriched UF_6 with the same dimensions as the 30B canister due to its criticality control system (CCS). The CCS consists of 33 natural B_4C rods organized in three concentric rings with 6, 12, and 14 rods respectively. In this study the CCS is modified to optimize the amount of HALEU fuel the 30B-10 can transport. This is done by studying the impact of the (CCR) radius, and their geometric layout, on the UF_6 mass while keeping k_{eff} (in unmoderated conditions) the same. The study calculates the rods worth to create a new optimized geometry and radius for the CCS so the canister can carry more U-235. The study is performed using Monte Carlo N-Particle Transport (MCNP)[1] code for its ability to perform complex criticality calculations. This study proposes alternative cylinder configurations that are more efficient in transporting more UF_6 mass, thus minimizing cost and time.

MCNP 30B-10 Model

The 30B-10 cylinder is modeled as a carbon steel cylinder with ellipsoid caps, straight skirt, and CCRs. The CCRs are modeled with a thin steel casing and the longitudinal stiffener/lattice holder is not modeled. The CCRs material is modeled with boron carbide (B_4C), with natural boron enrichment, and density 1.35g/cc. Nominal values were used from the Orano Safety Analysis Report[2]. The cylinder is filled with 99.5 wt% UF_6 and 0.5 wt% HF. The fuel is a High Assay Low Enriched Uranium (HALEU) with 10% enriched uranium-235. The outside of the cylinder is surrounded by air. There are two shortened rods located at the valves. It is important to note that only the 30B-10 CCRs are changed throughout the different trials. Figure 1 shows the modeled geometry.



(a) Vertical (b) Horizontal

FIG. 1: Cross sections of MCNP geometry used for the results of this paper. Color changes based on material. Yellow: steel casing of the cylinder. Magenta: 10% enriched UF_6 fuel. Blue: B_4C.

Optimization of the CCR Radius

By determining the CCR reactivity, the expected k_{eff} can be calculated without having to run MCNP. This helps in efficiently designing a new CCR which has infinite possibilities for rod size and location. All of the designs target to have a k_{eff} equal to 0.28494 with an uncertainty of 0.000472. This value is extracted from the Safety Analysis Report for the canister for an unmoderated condition[2]. For the cylinder to safely store the fuel, it is assumed that k_{eff} needs to be less than or equal to this k_{eff} value. The geometry studied herein removes every other rod in the third ring of the CCS as seen in Figure 2. This geometry is ideal for its low k_{eff} value and not changing the location of the B_4C rods. The third ring rods are chosen to be removed because they have the lowest worth meaning more rods can be removed. This allows for more volume removal, creating space for larger quantities of UF_6 and therefore U-235. However, to compensate for the small increase in reactivity by removing the rods, the radius of the remaining rods is increased.

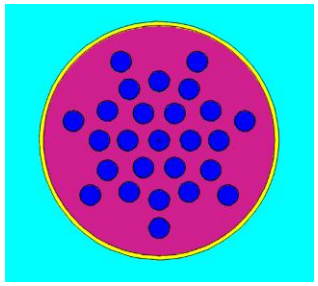


FIG 2. New CCS geometry with alternating rods removed from the third ring. Legend: Magenta=UF_6, Yellow=Steel, Blue=Natural B_4C, Teal=Air

The impact of changing the rod radius on k_{eff} is determined by running MCNP code. The results are shown in Table I.

TABLE I. Radius impact on k_{eff}

Radius (cm)	k_{eff}	Δk_{eff} Uncertainty
3.9596	0.228377	0.000405
3.4596	0.266609	0.000466
3.2596	0.283602	0.000464
3.2096	0.287518	0.000464
2.9596	0.312325	0.000487

To determine the optimized radius for the remaining rods a function of R (radius in centimeters) can be used to determine the mass of U-235 in the container. This is equation 1:

$$Mass[g] = 120640[g] - 26\left(54.17\left[\frac{g}{cm^2}\right] * (R[cm])^2\right) \qquad (1)$$

The first term 120 kg is the base mass of U-235 with no rods in the container. The mass of a single rod is the volume multiplied by density 1.34 g/cc. This mass is converted into UF_6 then into U-235 at 10% enrichment. This value is multiplied by 26 rods in the canister. The result from increasing the rod diameter on total U-235 mass is shown in Table II.

TABLE I. Radius impact on k_{eff}

Radius (cm)	U-235 (grams)
3.9596	98560.776
3.4596	103785.036
3.2596	105677.577
3.2096	106133.108
2.9596	108305.142

Using the new geometry, the value of k_{eff} from different rod radii is plotted. The results of the following data points are used to construct a linear fit to determine the optimized radius that will allow k_{eff} to be less than or equal to 0.28494. With no modifications the rod radius is 2.9596 cm and mass U-235 is 98,705.74 grams.

Threshold Values of the CCR Radius

The upper limit of the rod radii is restrained by the amount of uranium-235 that can be currently held in the cylinder. If the rods radii are increased beyond the threshold value, the volume left in the canister for UF_6 is not enough to increase the grams of U-235 above the current limit. This threshold is calculated by plotting radius and grams U-235 and restricting the value of the radius to only allow mass greater than 98,705.74 grams. From Figure 2, the max radius threshold is evaluated to be 3.947 cm.

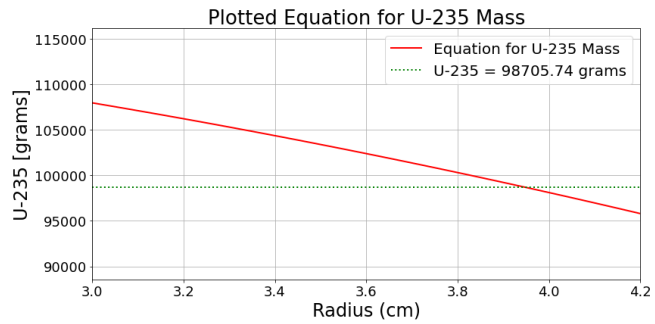


Fig. 2. A plot of radius vs grams of uranium 235 with equation 1 and a horizontal line at 98705.74 grams of uranium 235.

The lower limit of the rod radii is restrained by the value of k_{eff}. By plotting the data points for k_{eff} vs radius and fitting the data to a linear regression line, the value of the lower radii threshold can be calculated. As shown in Figure 3 and computed with equation 2, the lower radii threshold value is 3.23934 cm.

$$Keff = -0.0788\left[\frac{1}{cm}\right] * R[cm] + 0.5402 \qquad (2)$$

$$R^2(keff) = 0.9995$$

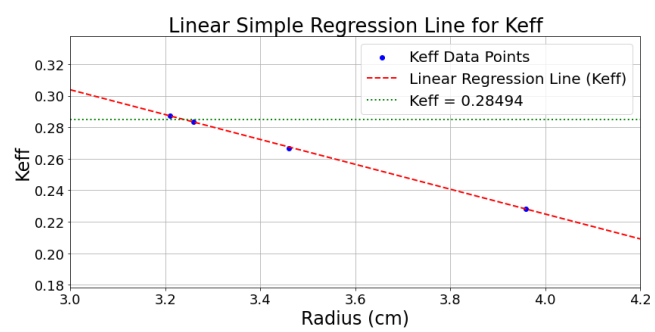


Fig. 3. A plot of radius vs k_{eff} with the k_{eff} data points from the computed simulations, a fitted with a linear fit and a horizontal line at k_{eff} equal to 0.28494.

RESULTS

The optimized design of removing alternating rods in the third ring of the CCR allows a range of 3.23934 cm to 3.947 cm. The results show that this design can carry more U-235 and have a lower k_{eff} than the current design. However, this analysis assumed that by keeping the same base k_{eff} value, the new geometry algins with the Safety Analysis Report by Orano and Urenco and meets the 10 CFR Part 71 NRC regulations.

Further studies are needed to verify the design is compatible with all requirements in 10 CFR Part 71 NRC regulations, especially in moderated conditions. A main concern is the addition of large vacancies from the removed rods in the third ring. The same space is a concern for large hydrogenated uranium residue (HUR) spheres to form in incident scenarios. Future work can be done to further optimize this design to remove the concerns listed above. More designs incorporating the second ring can also be analyzed. Further interesting analysis could incorporate artificial intelligence to help with optimization of design.

ACKNOWLEDGMENT

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2. M. Hennebach ET AL., “Safety Analysis Report for the DN30-X Package,” 0045-BSH-2020-001 Rev. 2, Orano (2022).